Std 12	**Lakshya**	**Marks :** 100
WEEKLY TEST	**SUBJECT** : Physics Fluid Mechanics, Surface tension and Viscosity	**Time (mm:ss) :** 60:00

Single Answer Correct 80

1. Two tubes of radii r_1 and r_2, and lengths l_1 and l_2, respectively, are connected in series and a liquid flows through each of them in steam line conditions P_1 & P_2 are pressure differences across the two tubes. If P_2 is $4P_1$ and l_2 is $\frac{\ell_1}{4}$, then the radius r_2 will be equal to : **4**

a. r_1

b. $2r_1$

c. $4r_1$

d. $\frac{r_1}{2}$

Ans : 4

Exp : The net flow rate remains same

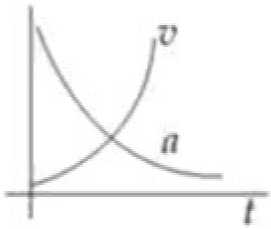
$$\therefore \quad Q = \frac{\pi P_1 r_1^4}{8\eta\ell_1} = \frac{\pi P_2 r_2^4}{8\eta\ell_2}$$

$$P_2 = 4P_1 \quad \ell_2 = \frac{\ell_1}{4}$$

$$\Rightarrow r_1^4 = 16 r_2^4$$

$$r_2 = \frac{r_1}{2}$$

2. Which of the following option correctly describes the variation of the speed v and acceleration '*a*' of a point mass falling vertically in a viscous medium that applies a force F =-kv, where '*k*' is a constant, on the body ? (Graphs are schematic and not drawn to scale) **4**

a.


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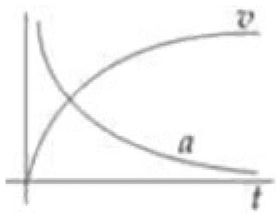

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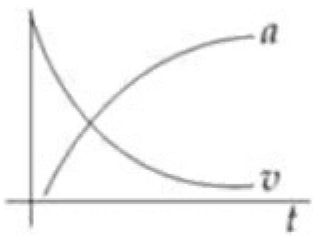
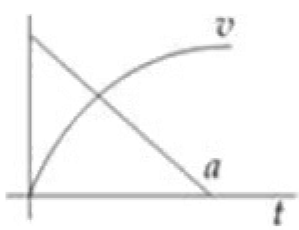


d.


Ans : 2

Exp : Force on the object

$\mathrm{F}_{\mathrm{Net}} = \mathrm{mg} - \mathrm{B} - \mathrm{F}_{\mathrm{R}}$

$a = g - \frac{\mathrm{B}}{m} - \left(\frac{6\pi\eta r}{m}\right)\mathrm{V}$

Hence a decreases as v increases

The dependence is exponential.

3. A large number of liquid drops each of radius r coalesce to from a single drop of radius R. The energy released in the process is converted into kinetic energy of the big drop so formed. The speed of the big drop is (given surface tension of liquid T, density ρ) : **4**

a. $\sqrt{\frac{6\mathrm{T}}{\rho}\left(\frac{1}{r}-\frac{1}{\mathrm{R}}\right)}$

b. $\sqrt{\frac{2\mathrm{T}}{\rho}\left(\frac{1}{r}-\frac{1}{\mathrm{R}}\right)}$

c. $\sqrt{\frac{\mathrm{T}}{\rho}\left(\frac{1}{r}-\frac{1}{\mathrm{R}}\right)}$

d. $\sqrt{\frac{4\mathrm{T}}{\rho}\left(\frac{1}{r}-\frac{1}{\mathrm{R}}\right)}$

Ans : 1

Exp : Let there be N drops. Then
Change in surface area $=\mathrm{N}\cdot 4\pi r^2-4\pi\mathrm{R}^2$
Energy released $=\mathrm{T}\left(4\pi\mathrm{N}r^2-4\pi\mathrm{R}^2\right)$
This energy goes into increasing the kinetic energy.
$\frac{1}{2}\,\mathrm{mV}^2=\mathrm{T.}\cdot 4\pi\left(\mathrm{N}r^2-\mathrm{R}^2\right)$
$\mathrm{V}=\sqrt{2r\cdot\frac{4\pi}{\mathrm{m}}\left(\mathrm{N}r^2-\mathrm{R}^2\right)}$
$=\sqrt{\frac{2\pi}{\rho}\cdot\frac{4\pi}{\frac{4}{3}\pi}\left(\frac{\mathrm{N}r^2}{\mathrm{R}^3}-\frac{\mathrm{R}^2}{\mathrm{R}^3}\right)}$
$\mathrm{V}=\sqrt{\frac{6\mathrm{T}}{\rho}\left(\frac{1}{r}-\frac{1}{\mathrm{R}}\right)}$

4. The velocity of water in a river is 18 km/hr near the surface. If the river is 5m deep, find the shearing stress between the horizontal layers of water. The co-efficient of viscosity of water = 10^{-2} poise. **4**

a. 10^{-4} N/m^2

b. 10^{-1} N/m^2

c. 10^{-2} N/m^2

d. 10^{-3} N/m^2

Ans : 4

Exp : Velocity of water at surface $=18\ \frac{\mathrm{km}}{\mathrm{hr}}=5\ \mathrm{m/s}$
Velocity of water in contact with river bed = 0
Depth of water = 5 m
Therefore, gradient $\frac{\Delta\mathrm{V}}{\Delta\mathrm{Z}}=15^{-1}$
Now stres $=\frac{\mathrm{F}}{\mathrm{A}}=\eta\left(\frac{\Delta\mathrm{V}}{\Delta\mathrm{Z}}\right)=10^{-2}\ \text{paise}\times 15^{-1}$
$=10^{-3}\ \mathrm{N}$

5. The average mass of rain drops is 3.0×10^{-5} kg and their average terminal velocity is 9 m/s. Calculate the energy transferred by rain to each square metre of the surface at a place which receives 100 cm of rain in a year. **4**

a. 4.05×10^4 J

b. 3.5×10^5 J

c. 9.0×10^4 J

d. 3.0×10^5 J

Ans : 1

Exp : No. of rain drops in 1 m^3 volume
$\mathrm{N} = \frac{1\ \mathrm{m}^3}{\text{Volume of 1 drop}}$
Assuming density of water $= 1000\ \frac{\mathrm{kg}}{\mathrm{m}^3}$
$\mathrm{V} = \frac{3 \times 10^{-5}}{1000} = 3 \times 10^{-8}\ \mathrm{m}^3$
$\therefore$ No of rain drops striking $= \frac{10^8}{3}$
Each rain drop carries energy $\mathrm{K} = \frac{1}{2}\ \mathrm{mV}^2$
$= \frac{1}{2} \times 3 \times 10^{-5} \times 9^2$
$\mathrm{K} = \frac{243}{2} \times 10^{-5}$ J
Hence total energy received $\mathrm{E} = \mathrm{NK}$
$= \frac{10^8}{3} \times \frac{243}{2} \times 10^{-5}$ J
$= 4.05 \times 10^4$ J

6. An air bubble of radius 0.1 cm is in a liquid having surface tension 0.06 N/m and density 10^3 kg/m^3. The pressure inside the bubble is 1100 Nm^{-2} greater than the atmospheric pressure. At what depth is the bubble below the surface of the liquid? (g = 9.8 ms^{-2}). **4**

a. 0.15 m

b. 0.1 m

c. 0.20 m

d. 0.25 m

Ans : 2

Exp : Pressure inside bubble $= \mathrm{P}_0 + d\rho g + \frac{2\mathrm{T}}{\mathrm{R}}$ at depth *d*.
Given $\frac{2\mathrm{T}}{\mathrm{R}} + d\rho g = 1100\ \mathrm{Nm}^{-2}$
$\frac{2 \times 0.06}{0.1 \times 10^{-2}} + d \times 10^3 \times 9.8 = 1100$
$d = 0.1$ m

7. Two soap bubbles coalesce to form a single bubble. If V is the subsequent change in volume of contained air and S the change in total surface area, T is the surface tension and P atmospheric pressure, which of the following relation is correct? **4**

a. 2PV + 3ST = 0

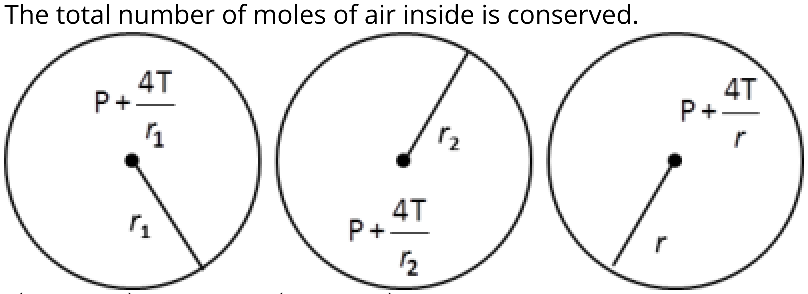
b. 3PV + 4ST = 0

c. 4PV + 3ST = 0

d. 3PV + 2ST = 0

Ans : 2

Exp : The total number of moles of air inside is conserved.



$$\left(\mathrm{P}+\frac{4\mathrm{T}}{r_1}\right)\frac{4}{3}\pi r_1^3+\left(\mathrm{P}+\frac{4\mathrm{T}}{r_2}\right)\frac{4}{3}\pi r_2^3=\left(\mathrm{P}\frac{4\mathrm{T}}{r}\right)\frac{4}{3}\pi r_3^3$$

$$\mathrm{P}\left(\frac{4}{3}\pi r^3-\frac{4}{3}r_1^3-\frac{4}{3}\pi r_2^3\right)+4\mathrm{T}\cdot\frac{4}{3}\pi\left(r^2-r_1^2-r_2^2\right)=0$$

$$\mathrm{PV}+\frac{4\mathrm{TS}}{3}=0\Rightarrow 3\mathrm{PV}+4\mathrm{TS}=0$$

8. A capillary tube is immersed vertically in water and the height of the water column is *x*. When this arrangement istaken into a mine of depth *d*, the height of the water column is *y*. If R is the radius of Earth, the ratio $\frac{x}{y}$ is : **4**

a. $\left(1-\frac{d}{\mathrm{R}}\right)$

b. $\left(1-\frac{2d}{\mathrm{R}}\right)$

c. $\left(\frac{\mathrm{R}-d}{\mathrm{R}+d}\right)$

d. $\left(\frac{\mathrm{R}+d}{\mathrm{R}-d}\right)$

Ans : 1

Exp : $h=\frac{2\mathrm{T}\cos\theta}{\rho g\mathrm{R}}$

Where T = surface tension

$\theta=$ angle of contact between capillary and water.

$\rho=$ density of water.

g = acc. due to gravity

R = radius of capillary

$h\propto\frac{1}{g}$(as others are constant)

$\Rightarrow\frac{x}{y}=\frac{g_{\text{at depend}}}{g_{\text{at surface of earth}}}$

$\Rightarrow\frac{x}{y}=\frac{g\left(1-\frac{d}{\mathrm{R}}\right)}{g}=\left(1-\frac{d}{\mathrm{R}}\right)$

9. Two small drops of mercury each of radius r from a single large drop. The ratio of surface energy before and after this change is : **4**

a. $2:2^{\mathbf{2/3}}$

b. $2^{\mathbf{2/3}}:2$

c. 2 : 1

d. 1 : 2

Ans : 1

Exp : If S is the surface tension, initial surface energy

$\mathrm{U}_i=\mathrm{S}\cdot4\pi r^2\times2=8\pi\mathrm{S}r^2$

When drops combine, the radius of bigger drop becomes

$\mathrm{R}=2^{\frac{1}{3}}r$

$\therefore\ \mathrm{U}_f=\mathrm{S}\cdot4\pi\mathrm{R}^2=4^{\frac{1}{3}}\cdot4\pi\mathrm{S}r^2$

$\frac{\mathrm{U}_i}{\mathrm{U}_f}=\frac{2}{2^{\frac{2}{3}}}$

10. Two capillaries of length L and 2L and of radius R and 2R are connected in series. The net rate of flow of fluid through them will be (given rate of the flow through single capillary, $\left(\mathrm{X}=\pi p \mathrm{R}^4 / 8 \eta \mathrm{L}\right)$ **4**

a. $\frac{8}{9}$

b. $\frac{9}{8}$

c. $\frac{5}{7}$

d. $\frac{7}{5}$

Ans : 1

Exp : Let P be the pressure head across the pipes, then

$$\frac{8 \cdot 3\mathrm{L}\ \eta}{\pi\ \mathrm{R}_{\mathrm{eq}}^4}=\frac{8 \cdot 2\mathrm{L} \cdot \eta}{\pi\ \mathrm{R}^4}+\frac{8 \cdot 2\mathrm{L} \cdot \eta}{\pi(2\mathrm{R})^4} \quad \Rightarrow \mathrm{R}_{\mathrm{eq}}=\left(\frac{8}{3}\right)^{\frac{1}{4}} \mathrm{R}$$

$$\therefore \quad \mathrm{Q}=\frac{\pi\ \mathrm{R}_{\mathrm{eq}}^4}{8 \eta \cdot 3\mathrm{L}} \mathrm{P}=\frac{8}{9} \frac{\pi \mathrm{R}^4 \mathrm{P}}{\eta \mathrm{L}}=\frac{8}{9}$$

11. The rate of flow of liquid in a tube of radius *r*, length ℓ, whose ends are maintained at a pressure difference *p* is $\mathrm{V}=\frac{\pi \mathrm{Q} p r^4}{\eta \ell}$, where η is coefficient of the viscosity and Q is : **4**

a. 8

b. 1/8

c. 16

d. 1/16

Ans : 2

Exp : Flow rate in a pipe is given by

$$\mathrm{V}=\frac{\pi \mathrm{P} r^4}{8 \eta \ell}$$

12. As the temperature of water increases, its viscosity **4**

a. Remains unchanged

b. Decreases

c. Increases

d. Increases or decreases depending on the external pressure

Ans : 2

Exp : viscosity decreases as temperature increases.

13. Water does not wet an oily glass, because **4**

a. Cohesive force of oil > adhesive force between oil and glass

b. Cohesive force of oil > cohesive force of water

c. Oil repels water

d. Cohesive force of water > adhesive force between water and oil molecules

Ans : 4

Exp : Force water-oil contact, the cohensive force between water molecues exceeds the adhesive force between water and oil molecules.

14. If work W is done in blowing a bubble of radius R from saop solution, then the work done in blowing a bubble of radius 2R from the same solution is **4**

a. $\frac{\mathrm{W}}{2}$

b. $2\mathrm{W}$

c. $4\mathrm{W}$

d. $2\frac{1}{3}\mathrm{W}$

Ans : 3

Exp : Work done goes into the surface energy of the bubble
$\mathrm{W} = 2 \cdot \rho \cdot 4\pi\mathrm{R}^2$
For bubble of radius 2R
$\mathrm{W}' = 2 \cdot \rho \cdot 4\pi(2\mathrm{R})^2 = 4\mathrm{W}$

15. One drop of water of diameter D breaks into 27 drops having surface tension T. The change in surface energy is **4**

a. $2\pi TD^{\mathbf{2}}$

b. $4\pi TD^{\mathbf{2}}$

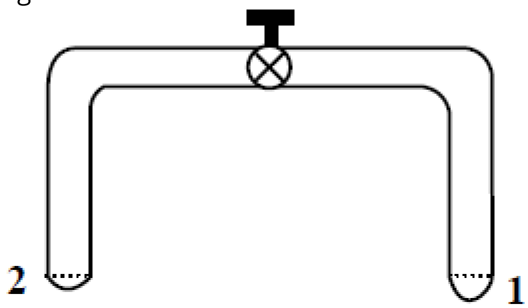
c. $\pi TD^{\mathbf{2}}$

d. $8\pi TD^{\mathbf{2}}$

Ans : 1

Exp : $W = T \times 4\pi R^{\mathbf{2}} \left(n^{\mathbf{1/3}} - 1\right)$
$W = 2\pi TD^{\mathbf{2}}$

16. A glass tube of uniform internal radius (r) has a valve separating the two identical ends. Initially, the valve is in a tightly closed position. End 1 has a hemispherical soap bubble of radius r. End 2 has sub-hemispherical soap bubble as shown in figure. Just after opening the valve. **4**
figure :



a. Air from end 1 flows towards end 2. No change in the volume of the soap bubbles

b. Air from end 1 flows towards end 2. Volume of the soap bubble at end 1 decreases.

c. No changes occurs

d. Air from end 2 flows towards end 1. Volume of the soap bubble at end 1 increases.

Ans : 2

Exp : P_1 = pressure just inside the bubble at the end $2 = P_0 + \frac{4T}{R}$
P_2 = pressure just inside the bubble at the end $1 = P_0 + \frac{4T}{r}$
$R > r \Rightarrow P_2 < P_1 \Rightarrow$ Air will flow from end 1 to end 2

17. **Statement I**: Tiny drops of liquid resist deforming forces better than bigger drops. **4**
Statement II: Excess pressure inside a drop is directly proportional to surface tension.

a. Statement I is true, Statement II is true and Statement II is a correct explanation for Statement I.

b. Statement I is true, Statement II is true and Statement II is NOT the correct explanation for Statement I.

c. Statement I is true, Statement II is false.

d. Statement I is false, Statement II is true.

Ans : 2

Exp : $\text{Excess pressure inside drop} = \frac{2\text{T}}{r}$
So if r is less $P_{\textbf{excess}}$ aull be more hence it will resist external deforming force.

18. A spherical solid ball of volume V is made of a material of density ρ_1. It is falling through a liquid of density ρ_2. $(\rho_2 < \rho_1)$.[Assume that the liquid applies a viscous force on the ball that is proportional to the square of its speedν,ie, F**viscous** = -kv**2**(k>0)] **4**

The terminal speed of the ball is

a. $\sqrt{\frac{Vg(\rho_1-\rho_2)}{k}}$

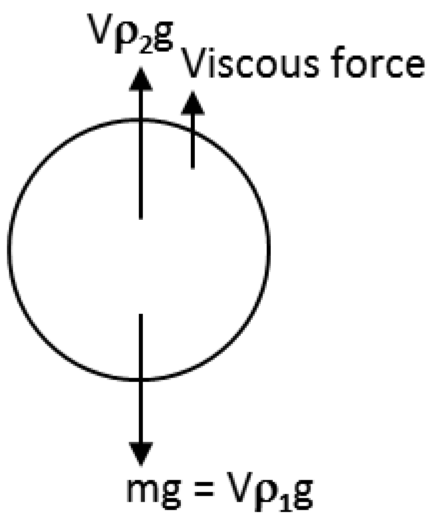
b. $\frac{Vg\rho_1}{k}$

c. $\sqrt{\frac{Vg\rho_1}{k}}$

d. $\frac{Vg(\rho_1-\rho_2)}{k}$

Ans : 1

Exp : The forces acting on the solid ball when it is falling through a liquid are mg downwards, thrust by Archimedes principle upwards and the force due to the force of friction also acting upwards. The viscous force of rapidly increases with velocity, attaining a maximum when the ball reaches the terminal velocity.
Then the acceleration is zero.



$mg - V\rho_2 g - k\nu^2 = ma$Where V is volume, v is the terminal velocity. When the ball is moving with terminal velocity a = 0.
Therefore $V\rho_1 g - V\rho_2 g - k\nu^2 = 0$
$\Rightarrow \nu\sqrt{\frac{Vg(\rho_1-\rho_2)}{k}}$

19. Work done in increasing the size of a soap bubble from radius of 3 cm to 5cm is nearly (surface tension of soap solution = 0.03 Nm^{-1}) **4**

a. $0.2\pi\, m\, J$

b. $2\pi\, m\, J$

c. $0.4\,\pi\, m\, J$

d. $4\,\pi\, m\, J$

Ans : 1

Exp : Work done in increasing radius of soap bubble
= U_f - U_i
$= 5\ 4\pi r_2^2 - 5.4\pi r_1^2$
$=\ 4\pi\ 5\,[r_2^2 - r_1^2]$
$= 0.2\pi mJ$

20. If a ball of steel (density $\rho = 7.8\ g\ cm^{-3}$) attains a terminal velocity of 10 cms-1 when falling in a tank of water (coefficient of viscosity $\eta_{\mathbf{water}} = 8.5 \times 10^{-4}\ \ Pa - s$) then its $terminal\ velocity\ in\ glycerine\ \left(\rho = 12\ \ g\ \ cm^{-3},\ \eta = 13.2Pa - s\right)\ would\ be\ nearly$ **4**

a. $1.6 \times 10^{-5}\, cms^{-1}$

b. $6.25 \times 10^{-4}\, cms^{-1}$

c. $6.45 \times 10^{-4}\, cms^{-1}$

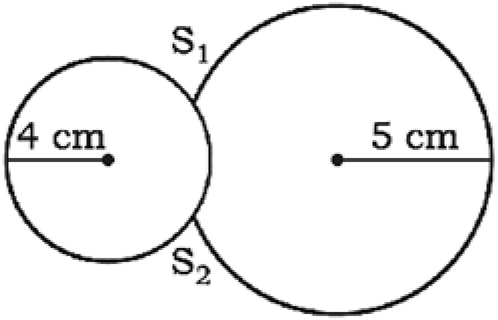
d. $1.5 \times 10^{-5}\, cms^{-1}$

Ans : 2

Exp : $\frac{V_b}{V_g} = \frac{13.2}{8.5\ \times\ 10^{-4}}$

Numerical Value Answer 20

1. Two soap bubbles of radii r_1 and r_2 equal to 4 cm and 5 cm are touching each other over a common surface S_1S_2 (shown in figure). Its radius will be **4**



Ans : 20

Exp : $r = \frac{r_1 r_2}{r_1 - r_2} = \frac{5 \times 4}{5-4} = 20\ cm.$

2. A small metal sphere of radius r and density ρ falls from rest in a viscous liquid of density σ and coefficient of viscosity η. Due to friction heat is produced. The rate of production of heat when the sphere has acquired the terminal velocity is proportional to r^n where n is **4**

Ans : 5

Exp : Rate of production of heat (= power) $= \mathrm{F_V . V_T}$

$= (6\pi\eta V_T r) \,.\, V_T$

$= 6\pi\eta r \left(V_{\mathbf{T}}^{\mathbf{2}}\right)$

$V_{\mathbf{T}} = \frac{2}{9} r^{\mathbf{2}} \frac{[\rho - \sigma] g}{\eta}$

$\therefore \quad \rho \alpha r^{\mathbf{5}}$

3. Consider two solid spheres P and Q each of density 8gm cm^{-3} and diameters 1 cm and 0.5 cm, respectively. Sphere P is dropped into a liquid of density 0.8 gm cm^{-3} and viscosity $\eta = 3$poiseulles. Sphere Q is dropped into a liquid of density 1.6 gm cm^{-3} and viscosity $\eta = 2$poiseulles. The ratio of the terminal velocities of P and Q is **4**

Ans : 3

Exp : $P_P = 8gm/cm^3 = P_Q$

$r_{\mathbf{P}} = 0.5 \times 10^{-\mathbf{2}} m$

$r_{\mathbf{Q}} = 0.25 \times 10^{-\mathbf{2}} m$

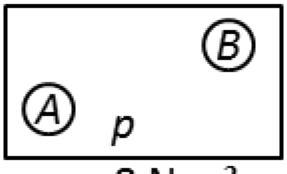
When sphere P is dropped into a liquid of density = 0.8 g/cm^3 and velocity $\eta_{\mathbf{p}} = 3$

$$\frac{\mathrm{V_P}}{\mathrm{V_Q}} = \frac{\frac{2}{9}\frac{r_{\mathbf{P}}^{\mathbf{2}} g}{\eta_{\mathbf{P}}}(\mathrm{P} - \sigma_{\mathbf{P}})}{\frac{2}{9}\frac{r_{\mathbf{Q}}^{\mathbf{2}} g}{\eta_{\mathbf{Q}}}(\mathrm{P} - \sigma_{\mathbf{Q}})} = \frac{r_{\mathbf{P}}^{\mathbf{2}}}{r_{\mathbf{Q}}^{\mathbf{2}}} \times \frac{\eta_{\mathbf{Q}}}{\eta_{\mathbf{P}}} \times \frac{(\mathrm{P} - \sigma_{\mathbf{P}})}{(\mathrm{P} - \sigma_{\mathbf{Q}})} = 3$$

4. Two soap bubbles A and B are kept in a closed chamber where the air is maintained at pressure $8Nm^{-2}$. The radii of bubbles A and B are 2cm and 4cm respectively. Surface tension of the soap-water used to make bubbles is 0.04 Nm^{-1}. Find the ratio $\frac{n_{\mathbf{B}}}{n_{\mathbf{A}}}$ where n_A and n_B are the number of moles of air in bubbles A and B, respectively. [Neglect the effect of gravity] **4**

Ans : 6

Exp : Although not given in the question, but we will have to assume that temperatures of A and B are same.



$p = 8\ Nm^{-2}$

$\frac{n_{\mathbf{B}}}{n_{\mathbf{A}}} = \frac{p_{\mathbf{B}} V_{\mathbf{B}}/RT}{p_{\mathbf{A}} V_{\mathbf{A}}/RT} = \frac{p_{\mathbf{B}} V_{\mathbf{B}}}{p_{\mathbf{A}} V_{\mathbf{A}}}$

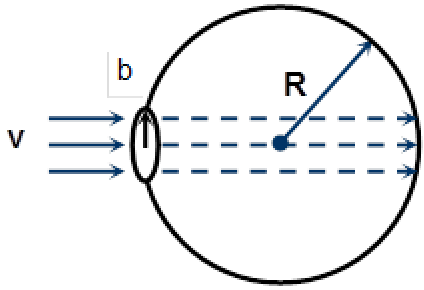
$= \frac{p + 4S/r_{\mathbf{A}} \times 4/3\pi\,(r_{\mathbf{A}})^{\mathbf{3}}}{(p + 4S/r_{\mathbf{B}}) \times 4/3\pi\,(r_{\mathbf{B}})^{\mathbf{3}}}$

(S = surface tension)

Substituting the values, we get

$\frac{n_{\mathbf{B}}}{n_{\mathbf{A}}} = 6$

5. A bubble having surface tension T and radius R is formed on a ring of radius b $(b<<R)$. Air is blown inside the tube with velocity v as shown. The air molecule collides perpendicularly with the wall of the bubble and stops. The radius at which the bubble separates from the ring is $\frac{\lambda T}{\rho v^2}$. Find λ. **4**



Ans : 4

Exp : $2\pi b \times 2T \sin\theta = \rho A v^2$

$\Rightarrow 4\pi b T \times \frac{b}{R} = \rho \pi b^2 v^2$

$\Rightarrow R = \frac{4T}{\pi v^2}$